

# Juneha Hwang

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🐙 <https://github.com/juneharold>

## Education

### Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

*Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)*

*Daejeon, South Korea*

- **Relevant Coursework:** Linear Algebra for Data Science (Python), Programming Principles (Rust), Data Structures (Java), Introduction to Algorithms, Machine Learning, Systems Programming (C), Programming Languages (Scala)

## Experience

### Linq Alpha

Jan 2025–Present

*Data Engineer*

*Seoul, South Korea*

- Architected a FastAPI-based API layer and asynchronous processing backend (Celery/Redis) to preprocess and deliver large public/licensed datasets with observability for long-running jobs.
- Managed large-scale financial datasets using PostgreSQL and Alembic, ensuring data integrity and performance.
- Engineered an entity resolution pipeline that maps ambiguous names (e.g. "MS", "Apple") to stock identifiers using a multi-stage system (cache → trigram search on DB → vector DB search → Decision-making using LLM).
- Designed and deployed a semantic document chunking system that transforms unstructured financial documents (earnings calls, broker research, internal investment notes, etc.) into structured queryable metadata, processing variable-format PDFs through a multi-batch LLM pipeline with multi-layer JSON parsing to handle non-deterministic LLM model outputs.

### FBA Quant (Quantitative Finance Research Group)

Feb 2024–Present

*Quant Developer Track*

*Seoul, South Korea*

- Developed expertise in asset pricing, portfolio management, and foundational financial engineering.
- Engineered an algorithmic trading system with the Interactive Brokers API, utilizing parallel processing and comprehensive backtesting in Python.

## Projects

### FABRIC Interpreter & Type Checker (KAIST CS320 Project #3) | Scala

- Implemented a Scala interpreter + static type checker for FABRIC, a functional language with polymorphic ADTs and pattern matching.
- Added a desugaring pass from concrete syntax to a simpler AST to reduce interpreter complexity.

### Facial Emotion Recognition Program | Keras, Tensorflow, Scikit-Learn

- Designed and implemented a Convolutional Neural Network (CNN) using Keras to classify facial emotions into seven distinct categories. Improved accuracy from 60% to 70% with hyperparameter-tuning.
- Developed a real-time application in Python integrating a live camera feed to identify and categorize facial emotions.

## Certificates and Honors

### Certificates

- Machine Learning with Python: A Practical Introduction, Deep Learning with Tensorflow, Deep Learning with Tensorflow (*IBM*)
- Using Python for Research (*HarvardX*)

### Honors

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025 Finalist
- Union of Clubs for Programming Contests (UCPC) 2024, 2025 Finalist
- Jane Street ETC 2024 1st place
- Fall 2023 Dean's List (KAIST)

## Technical Skills

**Programming Languages:** Python, C / C++, Rust, Scala, JavaScript, Typescript, Ruby, SQL, HTML/CSS

**Tech Stack:** FastAPI, PostgreSQL, Celery, Redis, React, Scikit-Learn, Numpy, Pandas, Git

**Languages:** Korean (native), English (native), Japanese (conversational)